Energy of the H 1s wavefunction

Calculate the kinetic energy and the potential energy for the H 1s wavefunction. What are the ratios between

a) kinetic and potential energy?

b) kinetic energy and total energy?

c) potential energy and total energy?

$$\psi_{1s}^{Z} = \sqrt{\frac{Z^3}{\pi a_0}}\, e^{-\frac{Zr}{a_0}}$$

$$\left(\frac{-\hbar^2}{2M_{Nuc}}\nabla^2 + \frac{-\hbar^2}{2m_e}\nabla^2 - \frac{1}{4\pi\epsilon_0}\frac{Ze\cdot e}{|\vec{R}_{Nuc} - \vec{r}_e|}\right)\psi(\vec{R}_{Nuc}, \vec{r}_e) = E\psi(\vec{R}_{Nuc}, \vec{r}_e)$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) \equiv \nabla^2$$

$$\rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(-\frac{Z}{a_0} r^2 e^{-\frac{Zr}{a_0}}\right) = \boxed{e^{-\frac{Zr}{a_0}}\left(\frac{Z^2}{a_0^2} - \frac{2Z}{a_0 r}\right)}$$

$$r^2\frac{\partial}{\partial r} \Rightarrow -\frac{Z}{a_0} r^2 e^{-\frac{Zr}{a_0}}$$

$$\hookrightarrow \frac{1}{r^2}\frac{\partial}{\partial r} \Rightarrow \left(-\frac{2Z}{a_0} r e^{-\frac{Zr}{a_0}} + r^2\frac{Z^2}{a_0^2} e^{-\frac{Zr}{a_0}}\right)\frac{1}{r^2}$$

$\rightarrow$ fixed nucleus

$$\hat{T} + \hat{V} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{1}{4\pi\epsilon_0}\frac{Ze\cdot e}{|\vec{r}_{el}|}$$

$$\hat{T}\psi = -\frac{\hbar^2}{2m_e}\sqrt{\frac{Z^3}{\pi a_0}}\left(\frac{Z^2}{a_0^2} - \frac{2Z}{a_0}\frac{1}{r}\right)e^{-\frac{Zr}{a_0}}$$

$$= -\frac{\hbar^2}{2m_e}\left(\frac{Z^2}{a_0^2} - \frac{2Z}{a_0}\frac{1}{r}\right)\psi$$

$$\rightarrow \hat{T} = -\frac{\hbar^2}{2m_e}\left(\frac{Z^2}{a_0^2} - \frac{2Z}{a_0}\frac{1}{r}\right) = -\frac{\hbar^2 Z^2}{2m_e a_0^2} - \frac{\hbar^2 Z}{2m_e a_0}\frac{1}{r} = -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2} + \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$$

$$\hat{V} = -\frac{Ze^2}{4\pi\epsilon_0}\frac{1}{r}$$

$$\frac{1}{a_0^2} = \frac{m_e^2 e^4}{16\pi^2\epsilon_0^2\hbar^4} \rightarrow \frac{\cancel{\hbar} Z^2}{2\cancel{m_e}}\frac{m_e^{\cancel{2}} e^4}{16\pi^2\epsilon_0^2\cancel{\hbar^4}} = -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2}$$

$$\hat{H} = \hat{T} + \hat{V} = -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2} + \frac{Ze^2}{4\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0}\frac{1}{r}$$

$$= -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2} \mathrel{\hat{=}} E$$

now easy:

$$\langle\hat{V}\rangle = -\int \frac{Z^3}{\pi a_0^3} e^{-\frac{2r}{a_0}} \frac{Ze^2}{4\pi\epsilon_0}\frac{1}{r} e^{-\frac{Zr}{a_0}} d^3r$$

$$= -\frac{Z^4 e^2}{4\pi^2\epsilon_0 a_0^3}\int \frac{1}{r} e^{-\frac{2Zr}{a_0}} d^3r$$

$$= -\frac{Z^4 e^2}{4\pi^2\epsilon_0 a_0^3}\int \frac{1}{r} e^{-\frac{2Zr}{a_0}} r^2 \sin\theta \, dr\, d\theta\, d\varphi \quad \text{0r 0π 02π}$$

$$= -\frac{Z^4 e^2}{4\pi^2\epsilon_0 a_0^3} 4\pi \int_0^\infty r e^{-\frac{2Zr}{a_0}} dr \longrightarrow \int_0^\infty x e^{-\xi x} dx = \frac{1}{\xi^2}$$

$$= -\frac{\cancel{Z^4} Z^2 e^2}{\pi\epsilon_0 \cancel{a_0^3}}\frac{\cancel{a_0^2}}{4\cancel{Z^2}} = -\frac{Z e^2}{4\pi\epsilon_0}\frac{Z}{a_0} = -\frac{Z^2 e^2}{4\pi\epsilon_0}\frac{m_e e^2}{4\pi\epsilon_0\hbar^2} = -\frac{m_e Z^2 e^4}{16\pi^2\epsilon_0^2\hbar^2}$$

$$\langle\hat{T}\rangle = -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2} + \left\langle\frac{Ze^2}{4\pi\epsilon_0}\frac{1}{r}\right\rangle = -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2} + \frac{Ze^2}{4\pi\epsilon_0}\left\langle\frac{1}{r}\right\rangle$$

we know $\frac{Z}{a_0}\frac{Ze^2}{4\pi\epsilon_0} = \frac{Z^2 e^2}{4\pi\epsilon_0}\frac{m_e e^2}{4\pi\epsilon_0\hbar^2} = \frac{Z^2 e^4 m_e}{16\pi^2\epsilon_0^2\hbar^2}$

$$= -\frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2} + \frac{m_e Z^2 e^4}{16\pi^2\epsilon_0^2\hbar^2} = \frac{m_e Z^2 e^4}{\pi^2\epsilon_0^2\hbar^2}\left(\frac{1}{16} - \frac{1}{32}\right) = \frac{m_e Z^2 e^4}{32\pi^2\epsilon_0^2\hbar^2}$$

$$\rightarrow T = -\frac{1}{2}V$$

$$\rightarrow T = -E$$

$$\rightarrow V = 2E$$